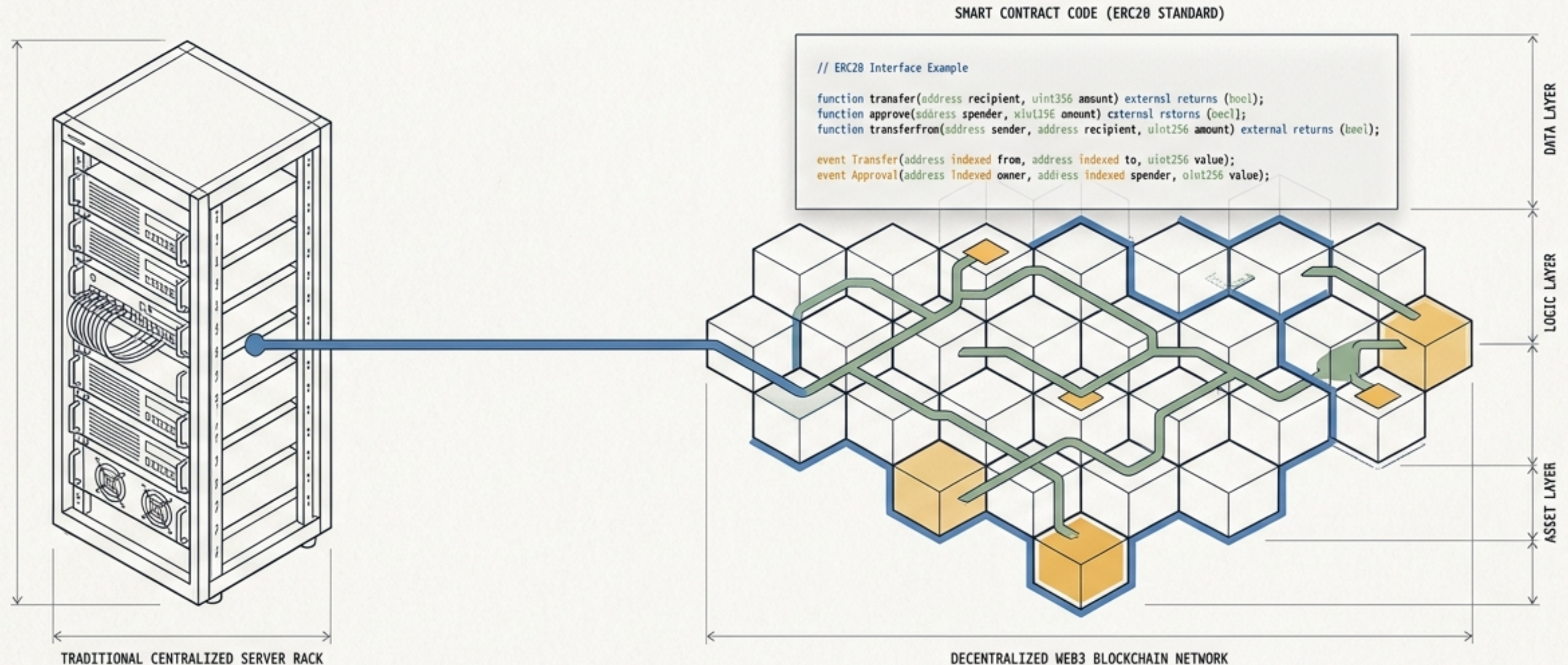


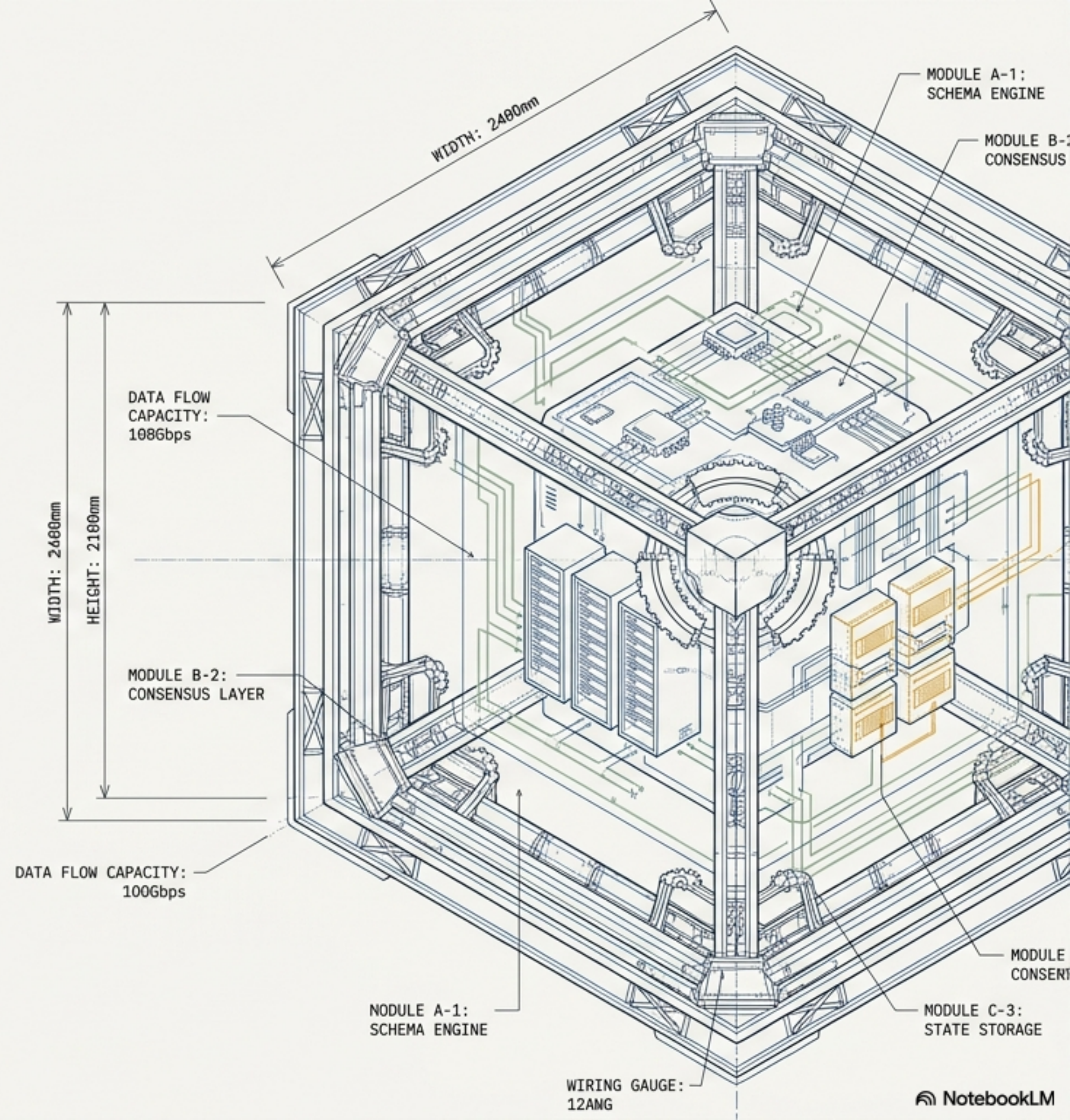
Demystifying Web3 Architecture

A Developer's Guide to Blockchains, Smart Contracts, and the ERC20 Standard



Part I: The Infrastructure Layer

Re-engineering the Database



The Blockchain as a Distributed State Machine

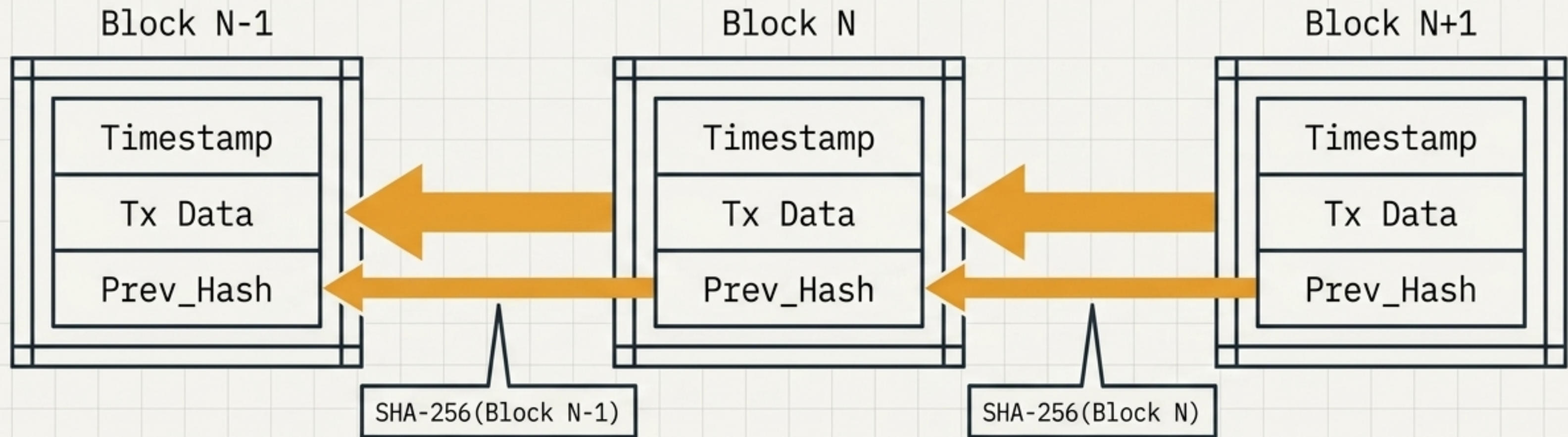
Trading centralized efficiency for decentralized trust.

	Traditional Client-Server IT	Blockchain / Web3 
Architecture	Centralized Database (SQL/NoSQL)	Highly Replicated Peer-to-Peer State Machine
Operations	C.R.U.D. (Create, Read, Update, Delete)	C.R. (Create, Read only)
Authority	Trusted Admin / Root Access	Cryptographic Consensus (No root admin)
Failure Point	Single point of failure	Distributed fault tolerance

BNB Chain processes over 3 billion transactions across 232 million unique addresses—acting as a massive, globally shared database requiring no centralized permission to access.

The Anatomy of Immutability

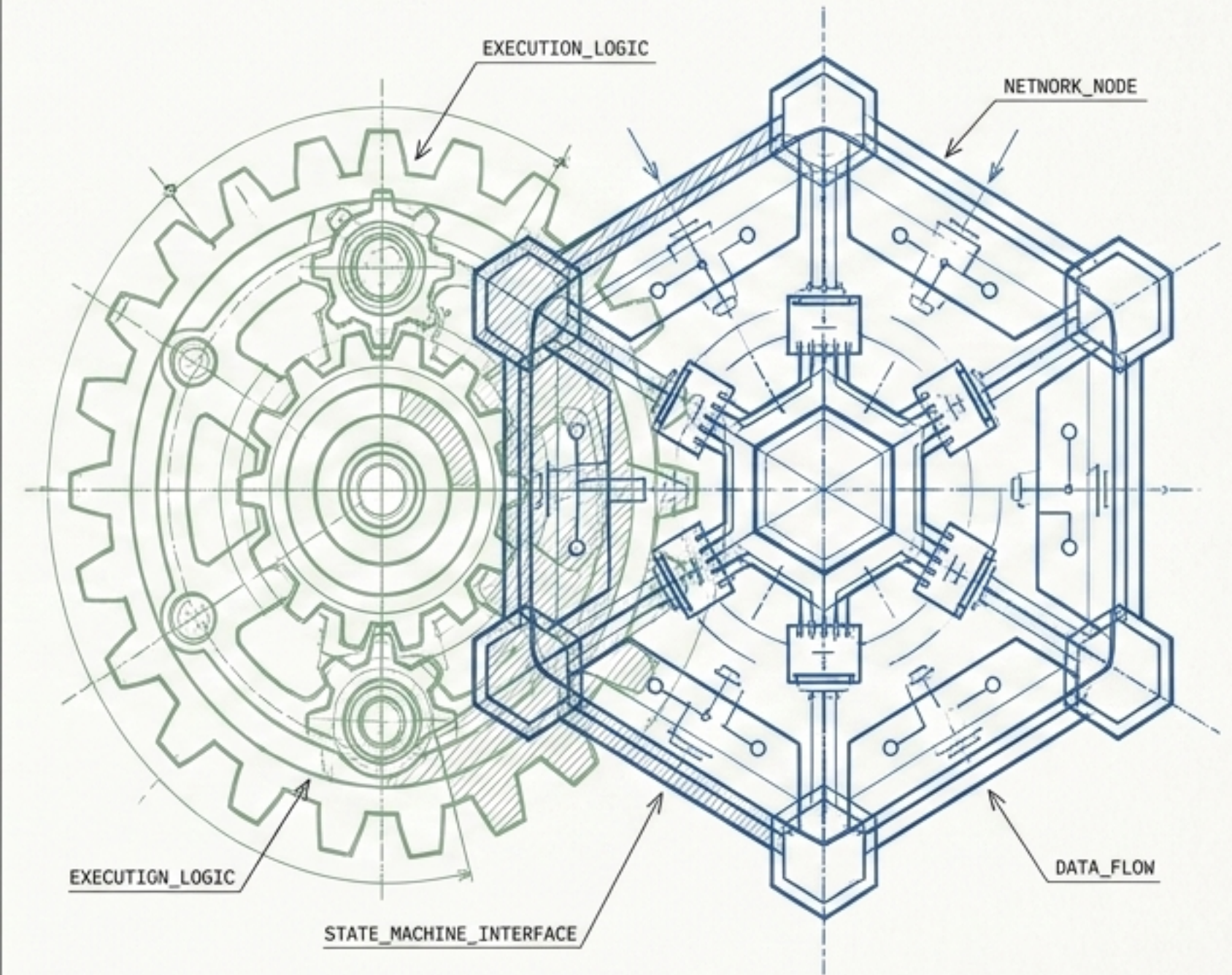
Append-only linked lists protected by cryptographic hashes.



Altering any historical transaction changes the payload. Changing the payload completely alters its resulting hash. A changed hash breaks all subsequent blocks in the chain. For a developer, a blockchain is simply an append-only linked list where the pointers are cryptographic hashes instead of memory addresses.

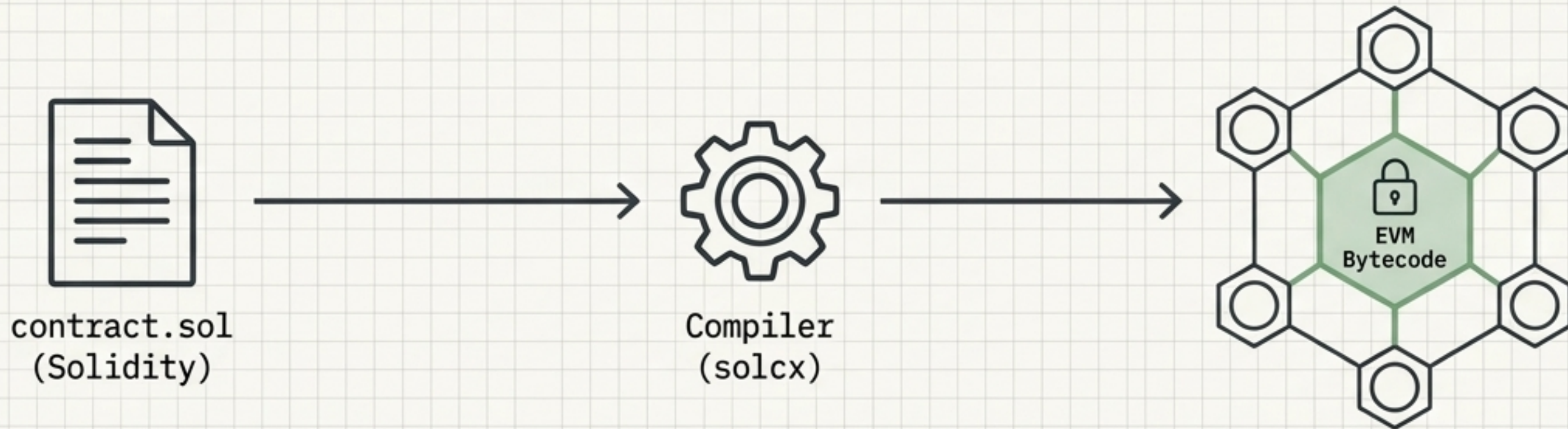
Part II: The Logic Layer

Programmable Execution
on the State Machine



The “Immutable Stored Procedure”

What exactly is a Smart Contract?



Backend Logic

It is self-executing code deployed directly to the blockchain.

No Server Maintenance

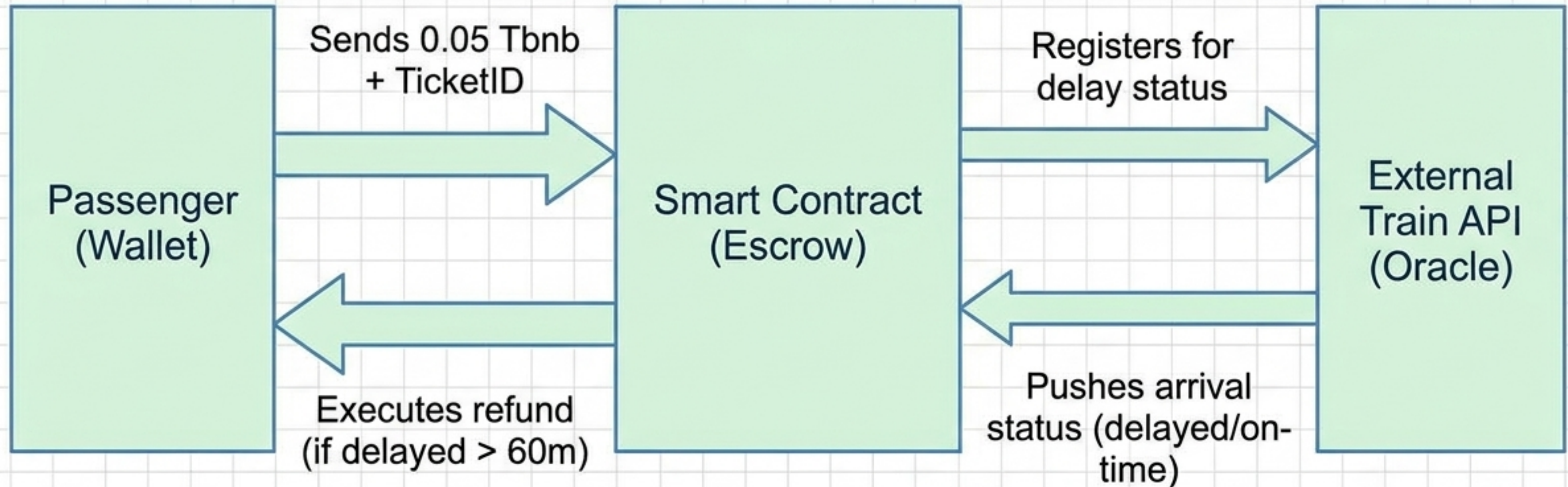
The network nodes execute the code; there is no AWS instance or localhost to maintain.

Immutable Execution

Once compiled and deployed, the code cannot be altered. It runs exactly as programmed, creating trustless execution.

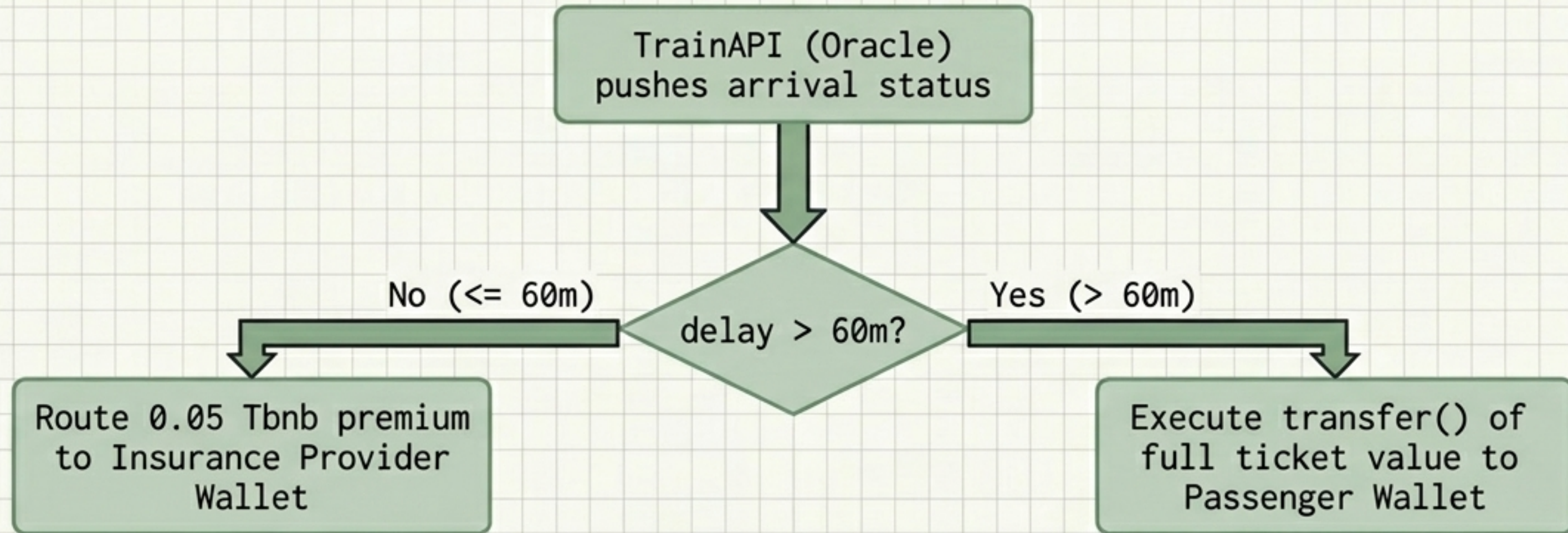
Architecture of Trust: Automated Train Insurance

Phase 1: Escrow and Conditional Logic Setup



Execution: Code as Law

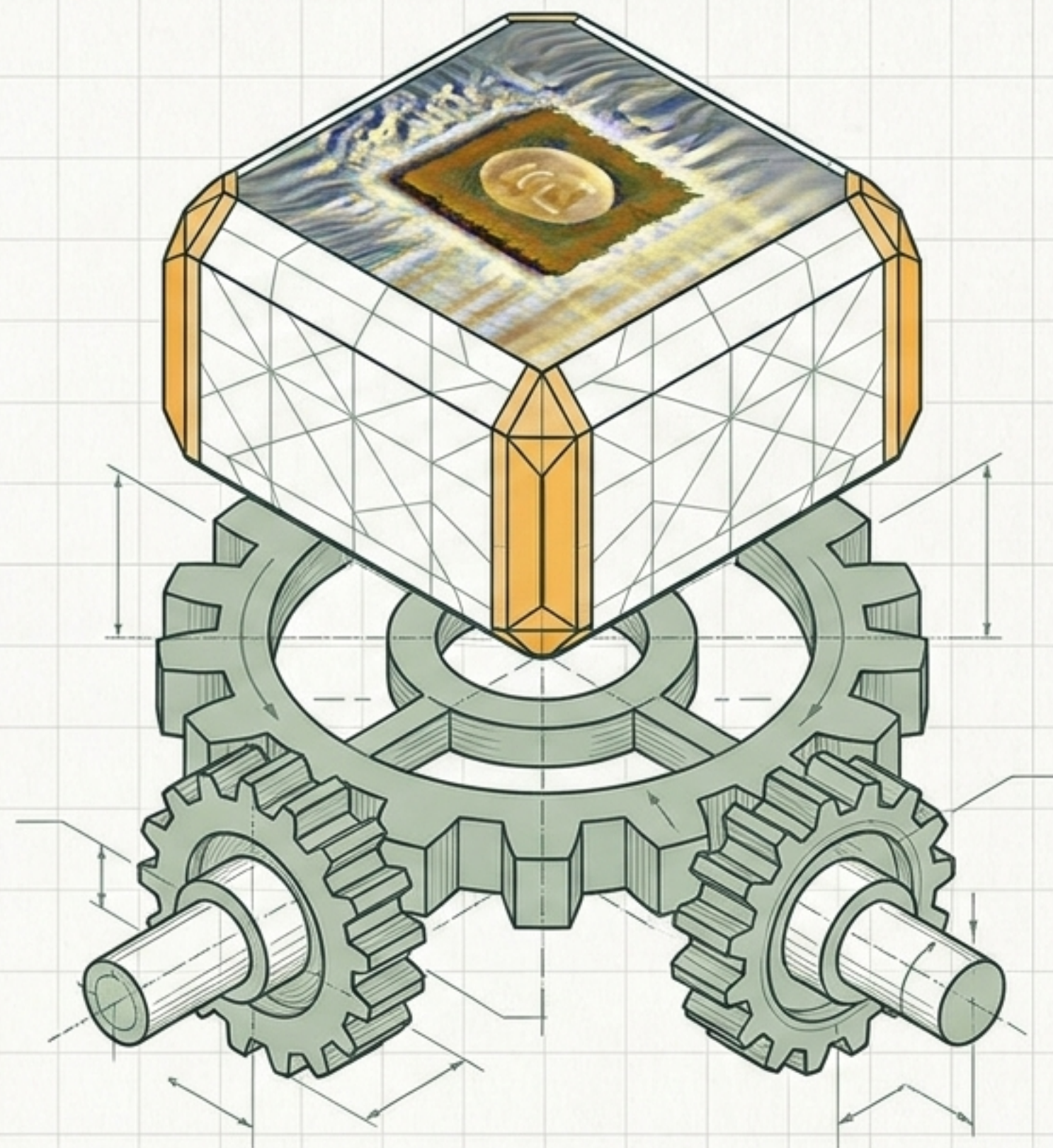
Phase 2: Oracle Trigger and Automated Settlement



Traditional insurance relies on centralized verification and manual payout processes. Smart contracts guarantee execution based strictly on pre-defined conditions. The code handles both the logic and the money movement simultaneously.

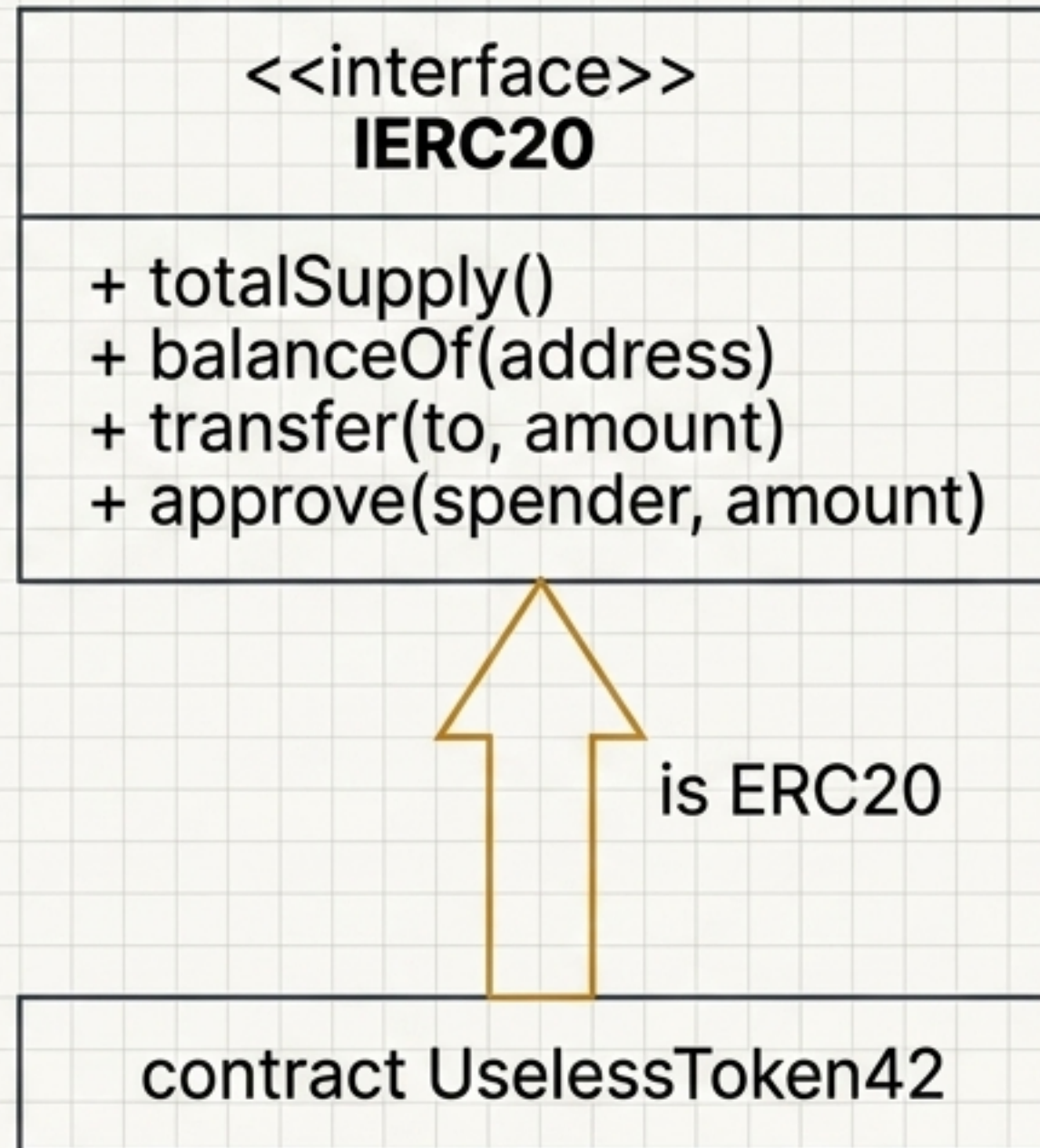
Part II: The Application Layer

Standardizing the Assets



ERC20: It's Just an API Standard

Tokens are not physical coins; they are standardized object classes.

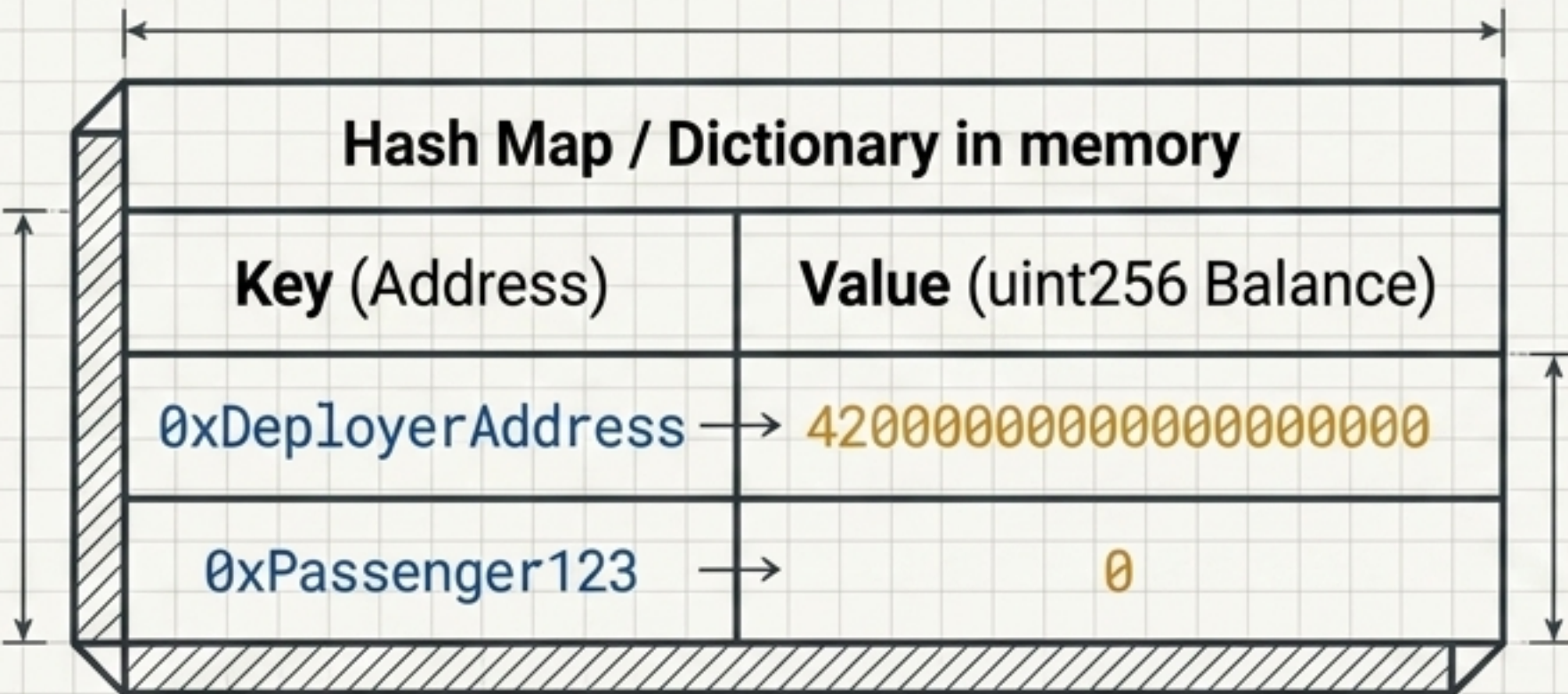


Instead of reinventing the wheel, developers inherit battle-tested, audited implementations.

By adhering strictly to the standard interface, external systems (wallets, exchanges) can interact with any newly minted token seamlessly.

Under the Hood: A Token is Just a Hash Map

Deconstructing UselessToken42



```
constructor()  
  ERC20("UselessToken42", "UT42") {  
    _mint(msg.sender, 42 * (10 ** 18));  
  }
```

Decimals	Transfers
Values are integers. The 10**18 multiplier accounts for 18 decimal places of divisibility, avoiding floating-point math entirely.	A token 'transfer' does not move a file. It simply calls a function that subtracts an integer from Sender A's dictionary key, and adds it to Receiver B's dictionary key.

The Web3 Architecture Layer Cake

How the ecosystem operates in unison.

